

IM YU A HABAROVA, Russian Federation

WMO: 217210

Lat: 72.400N

Lon: 126.350E

Elev: 64

StdP: 14.66

Time Zone: 9.00 (E09)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -42.4	(c) -38.9	(d) -46.7	(e) 0.4	(f) -42.3	(g) -43.2	(h) 0.5	(i) -38.6	(j) 50.4	(k) -18.8	(l) 45.0	(m) -21.4	(n) 5.3	(o) 180	(p) 1.114

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	71.3	60.7	65.8	57.3	60.7	54.4	61.4	69.9	58.0	64.9	54.8	60.3	11.9	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
57.1	69.7	66.4	54.2	62.6	62.8	51.2	56.0	58.9	27.5	70.5	25.2	64.4	23.2	60.3	71.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
41.0	36.4	32.1	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-44.5	77.7	4.5	6.9	-47.7	82.7	-50.4	86.7	-52.9	90.6	-56.2	95.6
			-44.6	64.2	4.2	3.1	-47.6	66.5	-50.1	68.3	-52.5	70.1	-55.6	72.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.0	-23.5	-20.9	-10.8	3.7	21.0	41.3	48.8	46.4	35.4	15.3	-9.6	-17.8		
	DBStd	28.13	8.52	10.89	13.78	12.43	8.92	10.32	8.35	6.50	6.81	10.25	10.54	10.38		
	HDD50	14423	2277	1986	1884	1388	898	304	122	154	442	1077	1789	2102		
	HDD65	19738	2742	2406	2349	1838	1363	714	511	578	889	1542	2239	2567		
	CDD50	171	0	0	0	0	0	42	85	41	3	0	0	0		
	CDD65	11	0	0	0	0	0	2	9	0	0	0	0	0		
	CDH74	91	0	0	0	0	0	18	73	0	0	0	0	0		
	CDH80	19	0	0	0	0	0	2	17	0	0	0	0	0		
	WSAvg	14.3	17.1	17.6	14.8	13.2	12.3	12.4	12.3	11.8	12.1	14.9	15.9	18.0		
	PrecAvg	16.2	1.1	0.7	0.8	0.6	1.2	1.5	2.2	2.4	1.7	1.6	1.1	1.3		
	PrecMax	23.9	2.8	2.6	1.8	2.3	2.5	2.8	5.0	5.0	4.2	2.6	2.1	3.3		
(15) Precipitation	PrecMin	11.3	0.3	0.1	0.2	0.0	0.3	0.4	0.7	1.0	0.7	0.5	0.4	0.2		
	PrecStd	2.9	0.6	0.6	0.4	0.5	0.6	0.7	1.0	0.9	0.8	0.6	0.5	0.8		
	DB	2.7	9.4	26.5	35.0	48.2	76.2	82.2	69.7	58.7	34.3	23.8	7.7			
	MCWB	2.0	8.5	25.3	32.8	41.8	61.2	67.8	60.6	52.7	33.3	22.9	6.7			
	2% DB	-2.9	3.2	22.5	29.8	41.4	69.1	74.3	64.4	50.7	32.5	17.6	3.5			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	MCWB	-3.3	2.6	21.5	28.0	37.7	58.7	62.4	57.7	47.7	31.4	16.6	2.8			
	5% DB	-7.4	-3.2	16.3	26.2	37.2	63.0	68.1	60.4	47.5	30.3	10.4	0.2			
	MCWB	-7.8	-3.9	15.4	24.3	34.5	54.5	59.2	54.8	44.5	29.2	9.8	-0.5			
	10% DB	-11.2	-6.6	11.7	22.8	34.2	57.0	62.6	57.2	44.3	28.6	5.2	-2.8			
	MCWB	-11.6	-7.1	10.8	21.2	32.3	51.3	56.1	52.9	41.8	27.7	4.5	-3.4			
	0.4% WB	2.0	8.6	25.6	32.4	42.3	62.6	68.2	61.3	52.5	33.8	23.3	6.9			
	MCDB	2.4	9.3	26.3	34.1	47.8	72.2	80.0	67.5	56.7	34.2	23.9	7.6			
	2% WB	-3.6	2.7	21.5	28.0	38.2	58.9	63.7	58.2	48.0	31.7	16.9	2.8			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	MCDB	-3.2	3.3	22.5	29.6	41.6	68.7	72.9	63.4	50.6	32.5	17.8	3.4			
	5% WB	-7.9	-3.7	15.4	24.5	34.6	55.2	59.7	55.3	44.9	29.3	9.9	-0.4			
	MCDB	-7.6	-3.2	16.3	26.0	36.7	62.0	67.1	59.5	47.2	30.1	10.6	0.2			
	10% WB	-11.6	-7.1	10.8	21.1	32.2	51.1	56.4	52.9	41.9	27.7	4.7	-3.3			
	MCDB	-11.3	-6.7	11.6	22.8	33.9	57.2	62.8	57.1	43.9	28.7	5.1	-2.8			
(35) Mean Daily Temperature Range	MDBR	8.5	8.7	11.2	12.6	8.9	10.6	10.9	9.5	6.5	7.2	8.5	8.7			
	5% DB	12.6	12.5	14.3	14.3	11.9	21.6	18.5	15.9	10.6	6.9	14.0	12.8			
	MCWBR	12.4	12.4	14.1	13.7	9.6	13.0	10.7	10.6	8.0	6.8	13.8	12.7			
	5% WB	12.7	12.4	14.4	14.0	11.1	20.8	18.3	14.4	10.0	6.9	13.9	12.7			
	MCWBR	12.4	12.3	14.3	13.5	9.4	13.4	11.8	10.5	8.2	6.9	13.7	12.6			
(40) Clear-Sky Solar Irradiance	taub	N/A	0.189	0.276	0.306	0.338	0.341	0.390	0.371	0.296	0.198	N/A	N/A			
	taud	N/A	1.945	2.042	2.088	2.202	2.442	2.320	2.337	2.418	1.977	N/A	N/A			
	Ebn at Noon	0	135	211	251	260	267	244	231	217	120	0	0			
	Edh at Noon	0	10	24	33	35	29	31	27	17	8	0	0			
(44) All-Sky Solar Radiation	RadAvg	0	58	504	1265	1715	1650	1400	883	431	102	0	0			
	RadStd	0	7	30	53	93	143	110	100	43	12	0	0			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)		+1.89	N/A	N/A	N/A	N/A	N/A	-743	-709	N/A	N/A

Nomenclature: See separate page